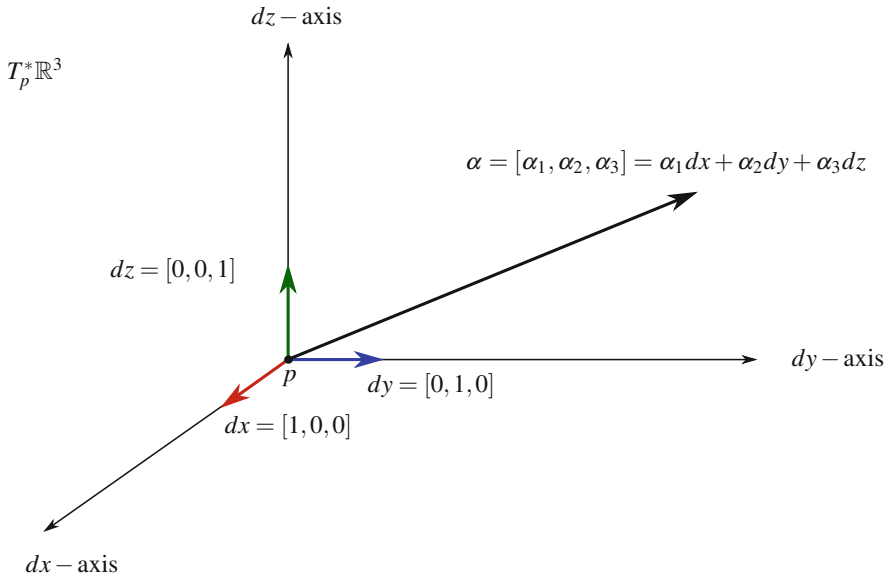
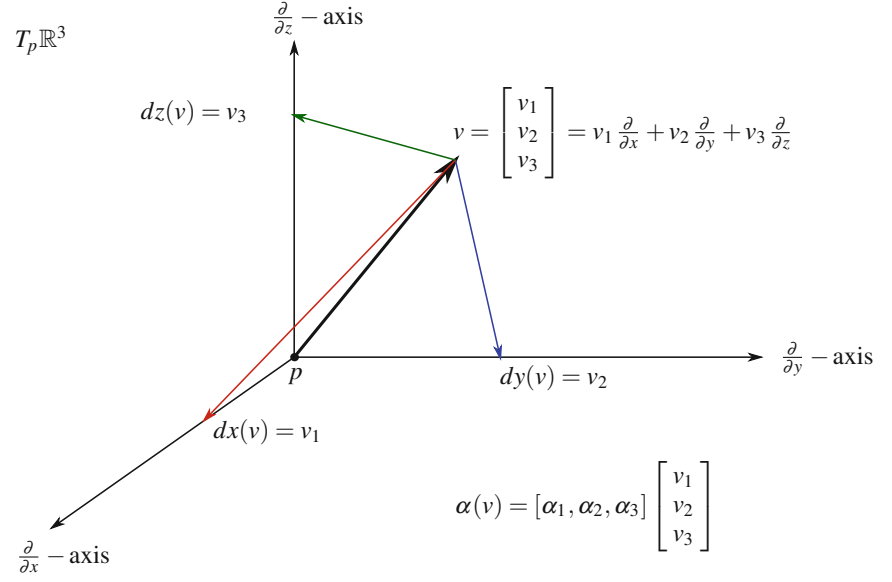

$dz-$axis
$T_p^*\mathbb{R}^3$
$\alpha=[\alpha_1,\alpha_2,\alpha_3]=\alpha_1 dx+\alpha_2 dy+\alpha_3 dz$
$dz=[0,0,1]$
$dy-$axis
p
$dy=[0,1,0]$
$dx=[1,0,0]$
$dx-$axis


$T_p\mathbb{R}^3$
$\frac{\partial}{\partial z}-$axis
$dz(v)=v_3$
$v=\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}=v_1\frac{\partial}{\partial x}+v_2\frac{\partial}{\partial y}+v_3\frac{\partial}{\partial z}$
$\frac{\partial}{\partial y}-$axis
p
$dy(v)=v_2$
$dx(v)=v_1$
$\alpha(v)=[\alpha_1,\alpha_2,\alpha_3]\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}$
$\frac{\partial}{\partial x}-$axis